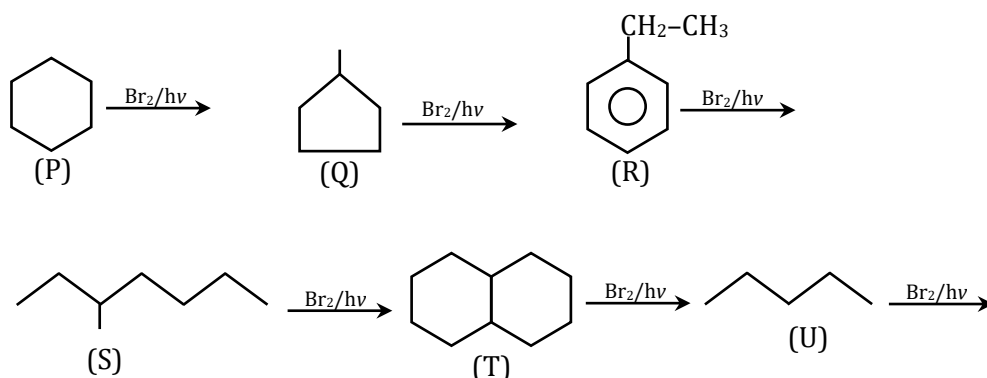




Reaction Mechanism : RACE-01

Carbocation, Freeradical substitutaion, Elimination Reactions

1. Among the following free radical bromination reactions, select those in which 2° halide is the major product.



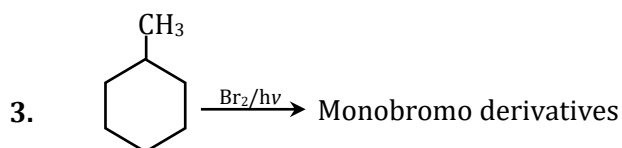
- (A) P, Q, R, S (B) P, R, U (C) P, R, S, T (D) P, Q, R, S, T

CRM132

2. Select the chain propagation steps in the free-radical chlorination of methane.

- (1) $\text{Cl}_2 \longrightarrow 2\text{Cl}^\bullet$ (2) $\text{Cl}^\bullet + \text{CH}_4 \longrightarrow \text{CH}_3\text{Cl} + \text{H}^\bullet$
 (3) $\text{Cl}^\bullet + \text{CH}_4 \longrightarrow \text{HCl} + \dot{\text{C}}\text{H}_3$ (4) $\text{H}^\bullet + \text{Cl}_2 \longrightarrow \text{HCl} + \text{Cl}^\bullet$
 (5) $\text{CH}_3^\bullet + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}^\bullet$
- (A) 2, 3, 5 (B) 1, 3, 6 (C) 3, 5 (D) 2, 3, 4

CRM133

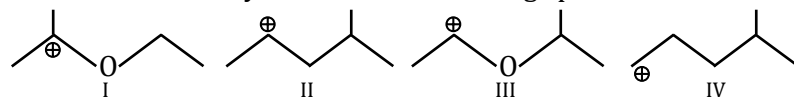


The number of possible monobromo products is (Excluding stereoisomerism)

- (A) 4 (B) 5 (C) 8 (D) 10

CRM134

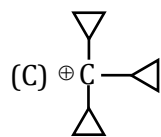
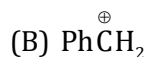
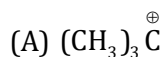
4. The correct stability order of the following species is :



- (A) II > IV > I > III (B) I > II > III > IV (C) II > I > IV > III (D) I > III > II > IV

CRM135

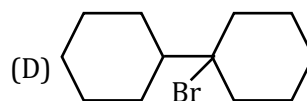
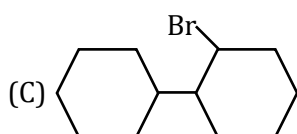
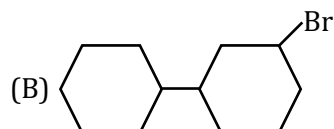
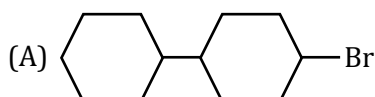
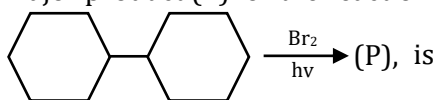
5. The more stable carbocation from the following is :



(D) All of these

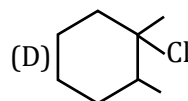
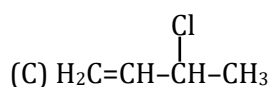
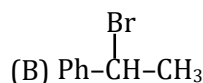
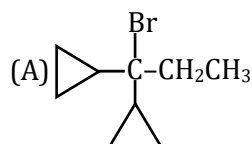
CRM136

6. Major product (P) for the reaction



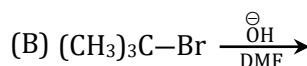
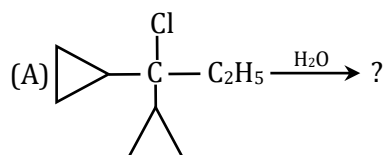
CRM137

7. Which of the following compounds will give E1 reaction?



CRM138

8. Which of the following will undergo E1 elimination?



(C) Both A and B

(D) None of these

CRM139

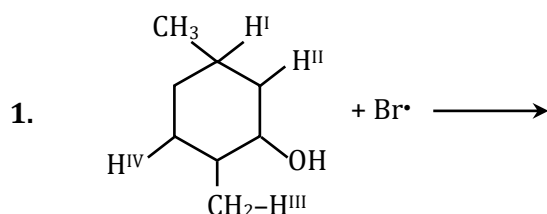
ANSWER KEY

Question	1	2	3	4	5	6	7	8	
Answer	B	C	B	D	C	D	ABCD	A	



Reaction Mechanism : RACE-02

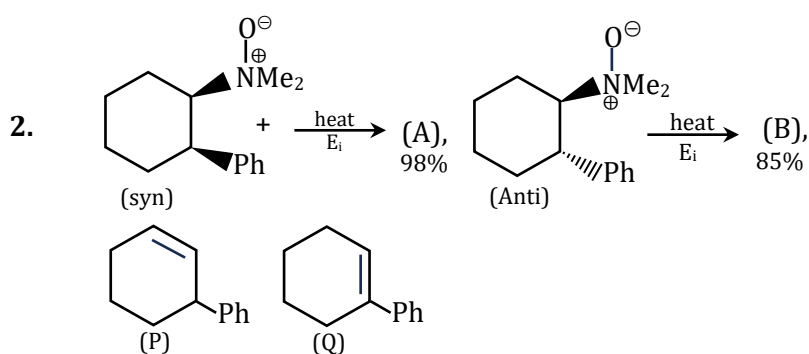
Freeradical substitutaion, Elimination Reactions



Br• will abstract which of the hydrogen most readily ?

- (A) I (B) II (C) III (D) IV

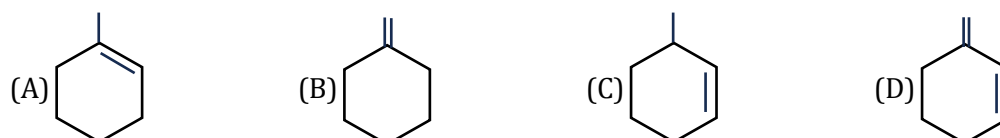
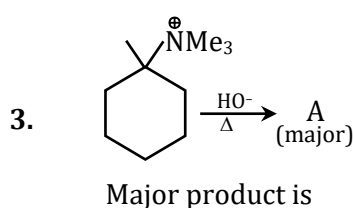
CRM140



Product (A) and (B) of the above reaction is

- (A) A = P, B = P (B) A = Q, B = Q (C) A = P, B = Q (D) A = Q, B = P

CRM141

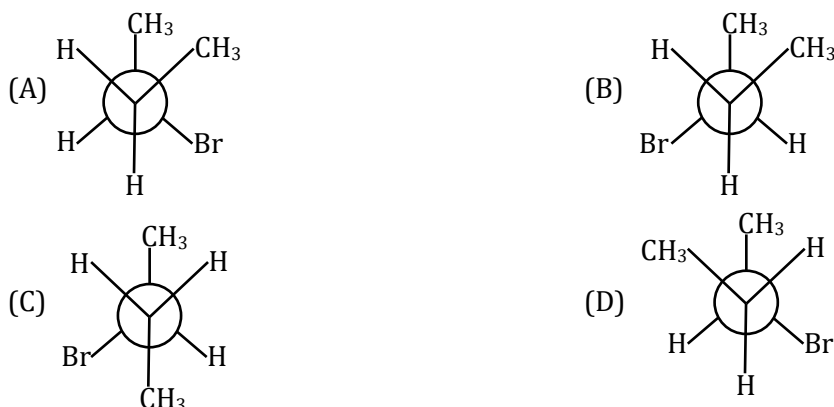


CRM142

4. The conversion of 2,3-dibromobutane to 2-butene with Zn is
 (A) Redox reaction (B) α-Elimination
 (C) β-Elimination (D) Both α-elimination and β-elimination

CRM143

5. In the dehydrohalogenation of 2-bromobutane, which conformations leads to the formation of cis-2-butene ?



CRM144

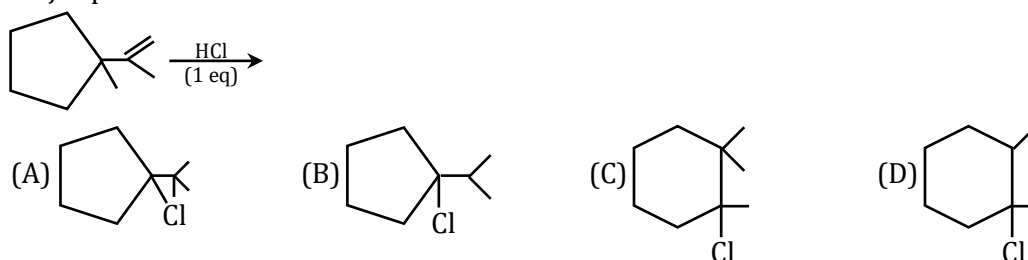
6. Match the items given in column I with those in column II.

	Column-I		Column-II
a.		p.	E1 (elimination unimolecular)
b.		q.	E2 (elimination bimolecular)
c.		r.	E1 cb (elimination from conjugate base)
d.		s.	Ei (Elimination intramolecular)

- (A) (a→r), (b→q), (c→p), (d→s)
 (B) (a→p), (b→q), (c→r), (d→s)
 (C) (a→s), (b→r), (c→q), (d→p)
 (D) (a→r), (b→q), (c→s), (d→p)

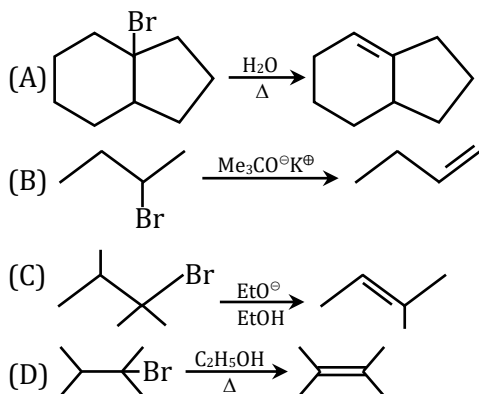
CRM145

7. Major product of the reaction is



CRM146

8. In which of the following cases, the major product has been correctly shown?



CRM147

9. Which of the following can give E₁ reaction?



CRM148

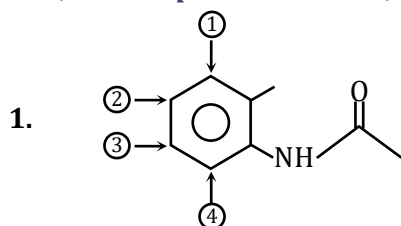
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	
Answer	A	C	B	C	A	A	C	BCD	AB	



Reaction Mechanism : RACE-03

ESR, Electrophilic Addition, Nucleophilic Substitution

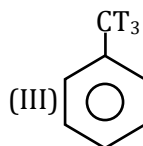
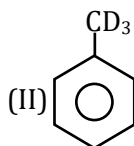
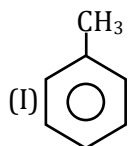


Sulphonation is the most favourable at the carbon number ...

- (A) 1 (B) 2 (C) 3 (D) 4

CRM149

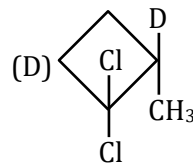
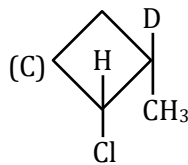
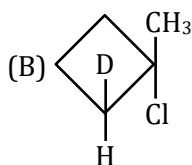
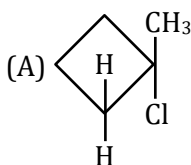
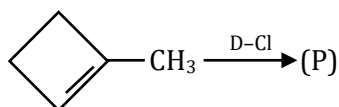
2. Which of the following shows highest reactivity towards EAS (electrophilic aromatic substitution).



- (A) 1 (B) 2 (C) 3 (D) 4

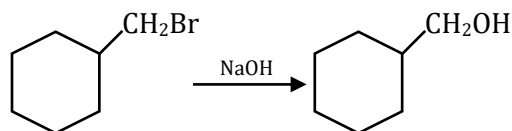
CRM150

3. Major product (P) in the following reaction is



CRM151

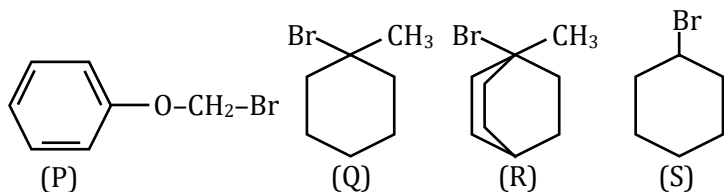
4. The suitable solvent for the reaction given below is



- (A) H_2O (B) $\text{C}_2\text{H}_5\text{OH}$ (C) HCONMe_2 (D) C_6H_6

CRM152

5. Rate of reaction of the given compounds with aqueous ethanol follows the order :

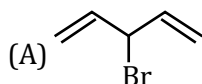


- (A) $P > Q > S > R$
 (C) $P > R > Q > S$

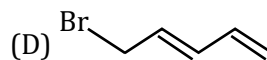
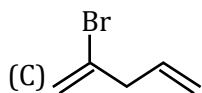
- (B) $Q > P > R > S$
 (D) $R > P > S > Q$

CRM153

6. Which of the following product will be formed by the reaction $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}_2 \xrightarrow[\text{hv}]{\text{NBS}} (\text{P})$?

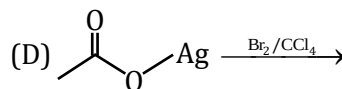
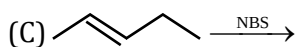
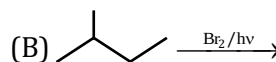
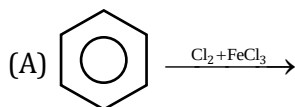


- (B) HBr

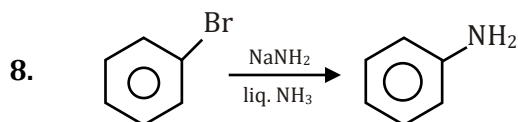


CRM154

7. Which of the following reactions involve free radical as intermediate ?



CRM155



The reaction is

- (A) $\text{ArS}_{\text{N}}1$
 (B) $\text{ArS}_{\text{N}}2$
 (C) Nucleophilic substitution via benzyne intermediate
 (D) None of these

CRM156

ANSWER KEY

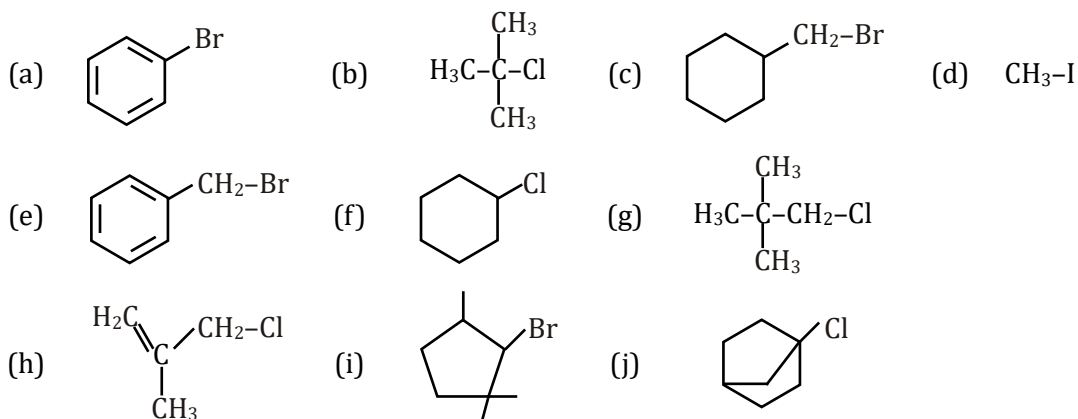
Question	1	2	3	4	5	6	7	8	
Answer	B	A	B	C	A	ABD	BCD	C	



Reaction Mechanism : RACE-04

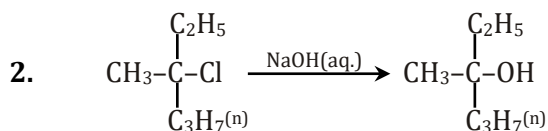
Nucleophilic Substitution, (S_N vs E)

1. Examine the ten structural formulas shown below and select that satisfy each of the following condition. Write one or more (a to j) in each of the question.



Which compound do not show S_N2 and Elimination reaction ?

CRM157

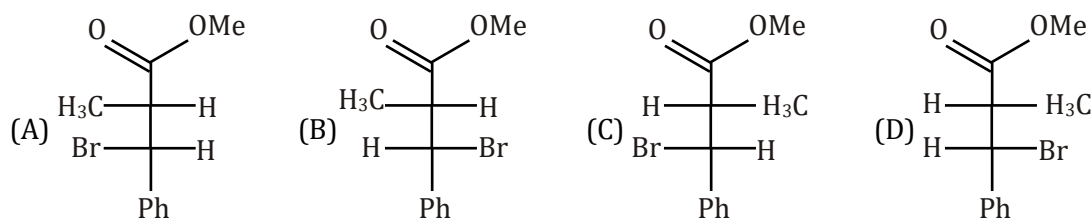
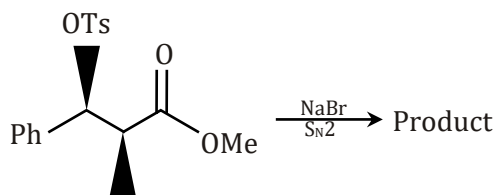


The product obtained will be

- (A) *d*- form (B) *l*- form
(C) racemic mixture mainly (D) none of these

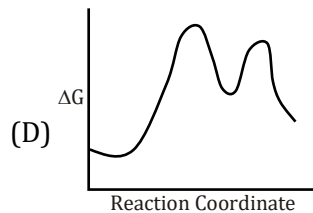
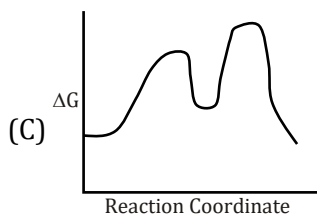
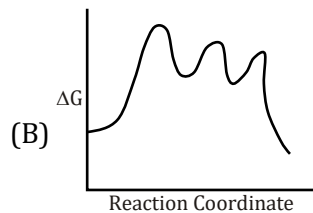
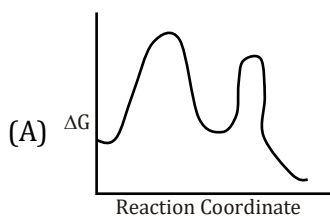
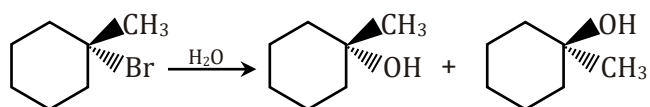
CRM158

3. What is the product of the following S_N2 reaction?



CRM159

4. Which of the correct reaction coordinate diagram for the following solvolysis reaction?



CRM160

5. Choose the correct among the following statements.

- (A) will react more readily than for S_N2 reaction
- (B) will react more readily than for S_N2 reaction
- (C) will react more readily than for S_N2 reaction
- (D) will react more readily than for S_N2 reaction.

CRM161

6. Consider the following statement and pick up the correct statement (s).

- (A) will react more readily than for S_N1 reaction.

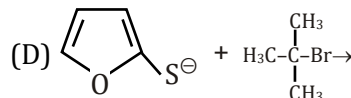
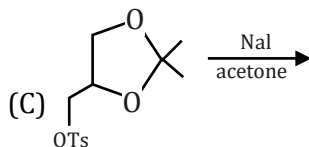
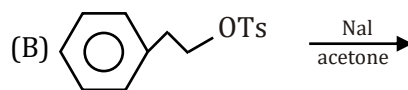
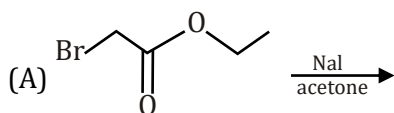
- (B) will react more readily than for S_N1 reaction.

- (C) will react more readily than for S_N1 reaction.

- (D) S_N1 reaction occurs in polar protic solvent.

CRM162

7. Which of the following reaction takes(s) place by S_N2 reaction?



CRM163

8. Which of the following statements is/are correct about the reactivities of (I) methyl chloride and (II) allyl chloride?

(A) Rate of S_N1 reaction (I) > (II).

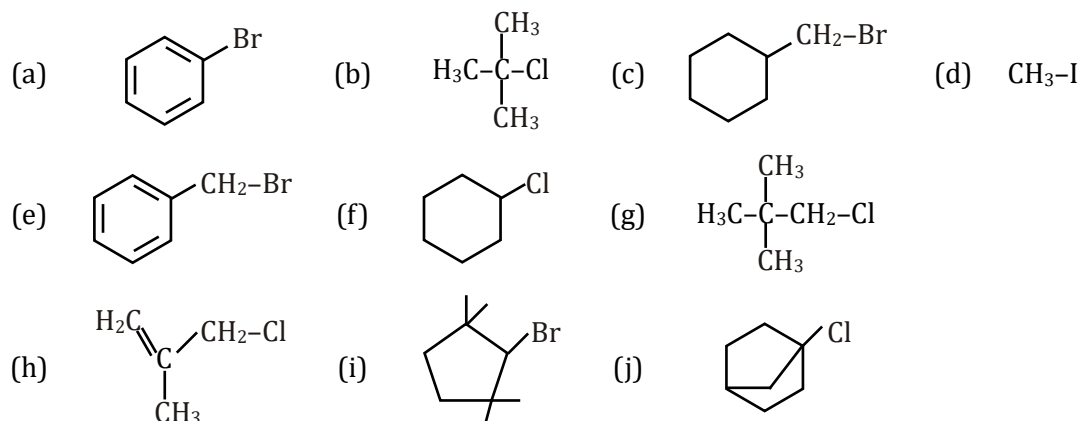
(B) Rate of S_N1 reaction (II) > (I).

(C) Rate of S_N2 reaction (I) > (II).

(D) Rate of S_N2 reaction (II) > (I).

CRM164

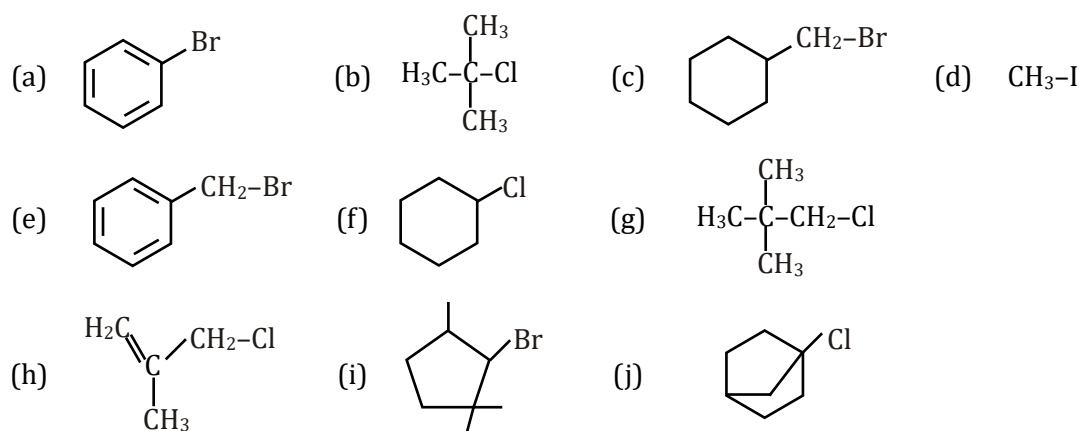
9. Examine the ten structural formulas shown below and select that satisfy each of the following condition. Write one or more (a to j) in each of the question.



Which compounds give an S_N2 substitution reaction on treatment with alcoholic NaSH?

CRM165

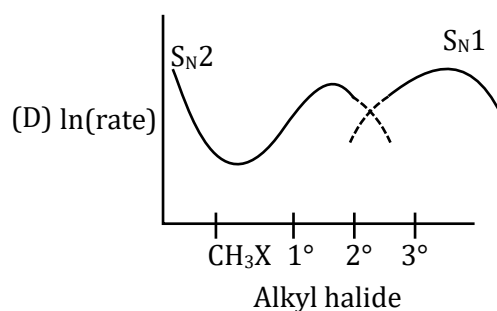
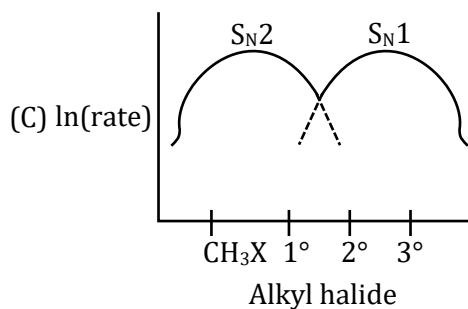
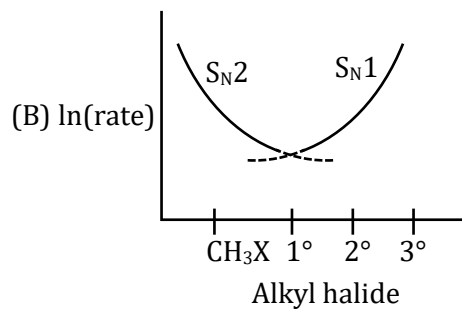
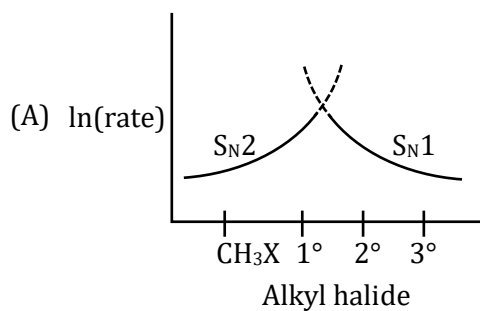
10. Examine the ten structural formulas shown below and select that satisfy each of the following condition. Write one or more (a to j) in each of the question.



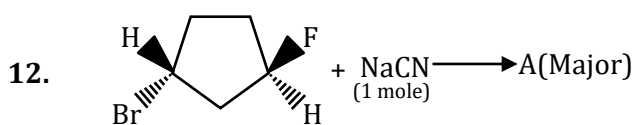
Which compounds give an $E2$ elimination reaction on treatment with alcoholic KOH?

CRM166

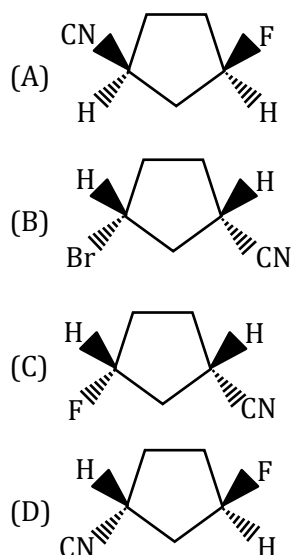
11. Which of the following curves correctly represents S_N1 vs S_N2 ?



CRM167



The major product of this reaction is/are



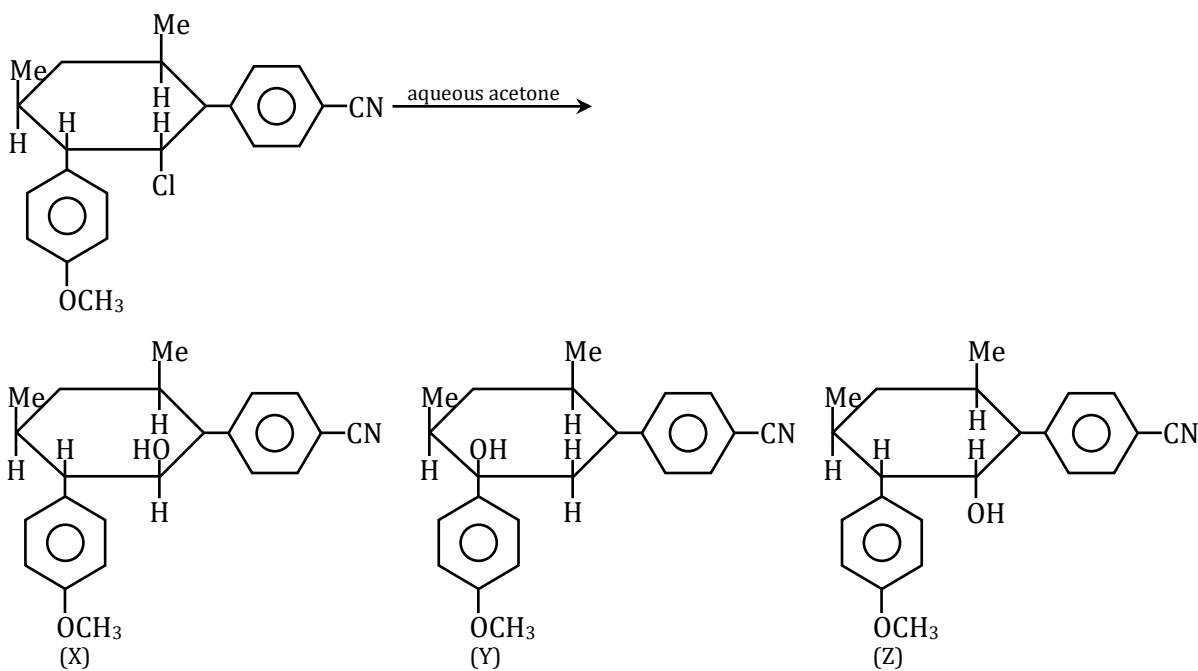
CRM168

13. The products of reaction of alcoholic silver nitrite with ethyl bromide are

- (A) Ethyne (B) Ethene
(C) Nitroethane (D) Ethyl nitrite

CRM169

14. Which of the following are correct regarding the products of the given reaction?



(A) only X and Z are formed
(C) $X > Y$ (amount)

(B) $Y > Z$ (amount)
(D) $Y > X$ (amount)

CRM170

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	3	C	A	A	ACD	ACD	ABC	BC	5	4
Question	11	12	13	14						
Answer	B	AC	CD	BD						